Lectures

MATH 107: Introduction to Linear Algebra — Fall 2026

Daily reading is listed with each meeting. Free Strang/MIT files and the Lachniet Octave PDF are collected on Resources.

Week 1

Mon Aug 24 — Why linear algebra? Vectors as representations

Lecture 01 page

Reading: Boyd 1.1

By the end of class, students should be able to:

1. Identify objects that can be represented as vectors.
2. Interpret vector entries, units, and indexing in context.
3. Distinguish scalars from vectors.
4. Translate a small data, position, signal, or measurement problem into vector form.

Wed Aug 26 — Vector arithmetic and geometry

Lecture 02 page

Reading: Boyd 1.2–1.3

By the end of class, students should be able to:

1. Add, subtract, and scale vectors.
2. Interpret those operations geometrically.
3. Plot vectors in $\mathbb{R}^2$ and recognize corresponding operations in $\mathbb{R}^3$.
4. Check whether an answer has sensible dimensions and units.

Fri Aug 28 — Linear combinations

Lecture 03 page

Reading: Linear combinations (class notes), Lab 0, and Lachniet Ch. 1 (Getting Started).

Note: No prior coding assumed. Lachniet is one PDF on Resources; chapters are listed here.

By the end of class, students should be able to:

1. Form and interpret linear combinations.
2. Explain the meaning of coefficients in context.
3. Enter, index, and modify vectors in Octave or Python.
4. Perform vector arithmetic and create a basic plot.
5. Modify supplied code and explain its output or error message.

Week 2

Mon Aug 31 — Inner product

Lecture 04 page

Reading: Boyd 1.4

By the end of class, students should be able to:

1. Compute a dot product.
2. Interpret it as a weighted sum.
3. Interpret its sign and magnitude as alignment.
4. Recognize orthogonal vectors from a zero dot product.

Wed Sep 2 — Linear and affine functions

Lecture 05 page

Reading: Boyd 2.1 and 2.3

Note: Boyd's calculus-dependent Taylor-approximation material is optional.

By the end of class, students should be able to:

1. Distinguish linear from affine functions in examples.
2. Test superposition numerically on simple functions.
3. Construct a function $a^\top x + b$ for a contextual quantity.
4. Evaluate a model by hand and computationally.

Fri Sep 4 — Linear functions in applications

Lecture 06 page

Reading: Applications of linear and affine models

By the end of class, students should be able to:

1. Model a score, cost, prediction, or aggregate quantity.
2. Interpret model coefficients and intercepts.
3. Compare outputs from competing models.
4. State a reasonable limitation or assumption of a linear model.

Week 3

Wed Sep 9 — Norm, distance, RMS, and variation

Lecture 07 page

Reading: Boyd 3.1–3.3

By the end of class, students should be able to:

1. Compute vector norms and distances.
2. Interpret norm and distance geometrically and contextually.
3. Relate norm to RMS and, where appropriate, variation around an average.
4. Use software to calculate distances in dimensions too large for hand computation.

Fri Sep 11 — Angle and similarity

Lecture 08 page

Reading: Boyd 3.4

By the end of class, students should be able to:

1. Compute cosine similarity or an angle between vectors.
2. Distinguish closeness in position from similarity in direction.
3. Choose an appropriate similarity measure for a stated purpose.
4. Explain why normalization may change a comparison.

Week 4

Mon Sep 14 — Nearest neighbors

Lecture 09 page

Reading: Boyd 4.1

Note: Possible contexts: species measurements, geographic sites, songs, or sensor readings.

By the end of class, students should be able to:

1. Formulate a nearest-neighbor question using vectors.
2. Find a nearest neighbor in a small dataset by hand.
3. Explain how feature scale can affect the answer.
4. Justify a choice of distance or similarity measure.

Wed Sep 16 — Clustering and k-means

Lecture 10 page

Reading: Boyd 4.2–4.3

Note: Use a small dataset students can visualize.

By the end of class, students should be able to:

1. Explain the goal of clustering.
2. Compute one assignment/update iteration of k-means.
3. Interpret a centroid as a vector.
4. Recognize that clustering output depends on representation, scaling, and initialization.

Fri Sep 18 — Finding similar things

Lecture 11 page

Reading: Lab 1 and Lachniet Ch. 1 (vectors, matrix storage, indexing, plots).

By the end of class, students should be able to:

1. Load and inspect a dataset represented by vectors.
2. Compute many distances with Octave or Python.
3. Adapt supplied code to identify nearest observations or clusters.
4. Validate computational output on a small hand-checkable subset.
5. Interpret results in the original context.

Week 5

Mon Sep 21 — Linear dependence as redundancy

Lecture 12 page

Reading: Boyd 5.1

By the end of class, students should be able to:

1. Recognize simple dependent and independent collections.
2. Construct a dependence relationship.
3. Interpret dependence as redundant information.
4. Explain why redundancy matters in data, measurement, and computation.

Wed Sep 23 — Span, independence, and subspaces

Lecture 13 page

Reading: Traditional vocabulary supplement to Boyd

By the end of class, students should be able to:

1. Define span using linear combinations.
2. Determine whether a vector lies in a simple span.
3. Interpret spans geometrically as lines, planes, or higher-dimensional collections.
4. Recognize basic subspaces and distinguish them from shifted lines or planes.

Fri Sep 25 — Basis, dimension, and orthonormal vectors

Lecture 14 page

Reading: Boyd 5.2–5.3

Note: Gram–Schmidt is a important application.

By the end of class, students should be able to:

1. Explain what a basis accomplishes.
2. Identify a basis and dimension in simple examples.
3. Use linear combination, span, subspace, independence, basis, and dimension accurately.
4. Recognize an orthonormal collection.
5. Explain why orthonormal coordinates are convenient for measurement and reconstruction.

Week 6

Mon Sep 28 — Unit I synthesis

Lecture 15 page

Reading: Unit I synthesis (no new Boyd section)

By the end of class, students should be able to:

1. Connect distance, similarity, span, and independence.
2. Select an appropriate vector tool for a new problem.
3. Move between algebraic, geometric, computational, and contextual representations.
4. Diagnose and explain common errors.

Wed Sep 30 — Midterm 1 review

Lecture 16 page

Reading: Unit I review

Note: Full review day.

By the end of class, students should be able to:

1. Retrieve major Unit I concepts without notes.
2. Solve mixed rather than chapter-isolated problems.
3. Explain reasoning collaboratively.
4. Identify gaps and make a targeted study plan.

Fri Oct 2 — Midterm 1

Lecture 17 page

Reading: Midterm 1 (in class)

By the end of class, students should be able to:

1. Compute with vectors and linear functions.
2. Interpret vector operations in context.
3. Analyze distance, similarity, span, and independence.
4. Select and communicate an appropriate method.

Week 7

Mon Oct 5 — Matrices as data and structured objects

Lecture 18 page

Reading: Boyd 6.1–6.3

By the end of class, students should be able to:

1. Identify matrix dimensions, rows, columns, and entries.
2. Represent datasets, networks, images, or systems as matrices.
3. Perform matrix addition, scaling, and transposition.
4. Recognize zero, identity, diagonal, triangular, symmetric, and sparse matrices.
5. Explain one practical consequence of matrix structure.

Wed Oct 7 — Matrix-vector multiplication

Lecture 19 page

Reading: Boyd 6.4

Note: Connect directly back to Week 1 linear combinations.

By the end of class, students should be able to:

1. Compute Ax by hand for a small matrix.
2. Interpret Ax as a linear combination of the columns of A.
3. Interpret A as an input-output transformation.
4. Explain how the dimensions of A determine the input and output spaces.

Fri Oct 9 — Transforming space

Lecture 20 page

Reading: Boyd 7.1–7.2, Lab 2, and Lachniet Ch. 2 (matrices and linear-algebra computations).

Note: Include a 3D or coordinate-system example.

By the end of class, students should be able to:

1. Predict the effect of scaling, reflection, shear, and rotation matrices.
2. Apply matrices to many 2D or 3D points at once.
3. Plot and compare original and transformed objects.
4. Adapt a short program that implements a transformation.
5. Interpret a matrix as a coordinate or physical-space transformation.

Week 8

Mon Oct 12 — Systems and $Ax = b$

Lecture 21 page

Reading: Boyd 8.3

By the end of class, students should be able to:

1. Translate equations or contextual constraints into $Ax = b$.
2. Interpret $Ax = b$ as asking whether b belongs to the span of the columns of A.
3. Classify systems as square, underdetermined, or overdetermined.
4. Anticipate circumstances producing zero, one, or many solutions.

Wed Oct 14 — Gaussian elimination

Lecture 22 page

Reading: Gaussian elimination (class notes) and Lachniet Ch. 2 (solving systems, RREF).

Note: Hand work first; software later.

By the end of class, students should be able to:

1. Perform elementary row operations.
2. Reduce a small augmented matrix to echelon form.
3. Solve a small system using back substitution.
4. Explain why row operations preserve the solution set.

Fri Oct 16 — RREF, pivots, free variables, and software

Lecture 23 page

Reading: RREF and software solvers; Lachniet Ch. 2 (linear-system commands).

Note: For Python, provide an RREF helper or use SymPy rather than making novices implement RREF.

By the end of class, students should be able to:

1. Interpret reduced row echelon form.
2. Identify pivot and free variables.
3. Describe a solution set parametrically.
4. Distinguish inconsistent, unique, and infinite-solution cases.

5. Use software to solve or row-reduce a system and verify the answer using a residual.

Week 9

Mon Oct 19 — Linear dynamical systems

Lecture 24 page

Reading: Boyd 9.1–9.3

By the end of class, students should be able to:

1. Represent an evolving system as $x_{t+1} = Ax_t$.
2. Interpret the state vector and transition matrix.
3. Build a transition matrix from a flow description.
4. Compute and interpret the next few states.

Wed Oct 21 — Simulation and long-run behavior

Lecture 25 page

Reading: Simulation of linear dynamics

Note: Do not explain the eigenvalue mechanism yet; let students form conjectures.

By the end of class, students should be able to:

1. Simulate many time steps computationally.
2. Plot state components through time.
3. Recognize stabilization, growth, decay, or oscillation.
4. Conduct a simple what-if simulation.
5. State limitations of a linear dynamical model.

Fri Oct 23 — Predict a system through time

Lecture 26 page

Reading: Lab 3, Lachniet Ch. 2 (matrix operations), and Lachniet's introductory programming/loops material.

Note: This creates the eigenvalue "mystery."

By the end of class, students should be able to:

1. Implement a linear dynamical model.
2. Write or complete a short iterative calculation.
3. Validate the first steps by hand.
4. Compare results under altered parameters or initial states.
5. Record an unexplained long-run pattern for later investigation.

Week 10

Mon Oct 26 — Matrix multiplication

Lecture 27 page

Reading: Boyd 10.1

By the end of class, students should be able to:

1. Determine whether a product is defined and give its dimensions.
2. Compute a small matrix product.
3. Interpret ABx as a sequence of transformations.
4. Explain why matrix multiplication is generally not commutative.

Wed Oct 28 — Composition and matrix powers

Lecture 28 page

Reading: Boyd 10.2–10.3

Note: Return to the dynamics lab from a new viewpoint.

By the end of class, students should be able to:

1. Combine multistage processes using a matrix product.
2. Connect $A^k x$ to repeated dynamics.
3. Interpret selected entries of a matrix product or power.
4. Recognize how network and transition information accumulates through powers.

Fri Oct 30 — Same mathematics, very different computational cost

Lecture 29 page

Reading: Boyd's $ab^\top c$ example (micro-lab)

Note: Precursor to the capstone.

By the end of class, students should be able to:

1. Compare algebraically equivalent orders of computation.
2. Count intermediate values that must be stored.
3. Measure runtime for increasing problem sizes.
4. Explain why forming a large intermediate matrix may be wasteful.
5. Choose a computational order and justify it quantitatively.

Week 11

Mon Nov 2 — Inverses and solving

Lecture 30 page

Reading: Boyd 11.1–11.3

By the end of class, students should be able to:

1. Interpret an inverse as undoing a transformation.
2. Determine whether a simple matrix is invertible.
3. Connect invertibility to uniqueness of $Ax = b$.

4. Explain why solving $Ax = b$ is generally preferable computationally to explicitly calculating A^{-1}.

Wed Nov 4 — Rank, column space, null space, and basis connections

Lecture 31 page

Reading: Boyd plus selected portions of Strang §3.5, especially rank, column space, null space, basis, and dimension.

Note: Not a full four-subspaces theory unit. Row space and left nullspace are enrichment, not quiz material.

By the end of class, students should be able to:

1. Interpret column space and null space through $Ax = b$ and $Ax = 0$.
2. Determine rank from pivots or independent columns.
3. Find simple bases for column or null spaces.
4. Connect rank to consistency, independence, and dimension.
5. Use subspace, basis, dimension, rank, column space, and null space correctly.

Fri Nov 6 — Determinants, geometry, and invertibility

Lecture 32 page

Reading: Determinants (class notes)

By the end of class, students should be able to:

1. Compute determinants of 2×2 and simple triangular matrices.
2. Interpret determinant magnitude as area or volume scaling.
3. Connect determinant zero with lost dimension and noninvertibility.
4. Recognize the determinant of a triangular matrix from its diagonal.
5. Connect determinants to the coming characteristic equation.

Week 12

Mon Nov 9 — Midterm 2 review and synthesis

Lecture 33 page

Reading: Unit II review

Note: Full review day.

By the end of class, students should be able to:

1. Connect transformations, systems, dynamics, rank, inverses, and determinants.
2. Compare hand and computational solution methods.
3. Interpret software output instead of merely reporting it.
4. Solve a new application requiring more than one concept.

Fri Nov 13 — Midterm 2

Lecture 34 page

Reading: Midterm 2 (in class)

By the end of class, students should be able to:

1. Analyze matrices as transformations and models.
2. Solve and interpret systems.
3. Use matrix products, powers, rank, inverses, and determinants appropriately.
4. Interpret computational results in context.

Week 13

Mon Nov 16 — Eigenvectors as persistent directions

Lecture 35 page

Reading: Beginning of Strang §6.1 — $Ax = \lambda x$, eigenvectors, and dynamics.

Note: Start with the unexplained behavior from Lab 3. The free PDF is only §6.1.

By the end of class, students should be able to:

1. Explain geometrically what makes v an eigenvector.
2. Verify a proposed eigenpair.
3. Find λ when an eigenvector is given.
4. Connect eigenvectors to repeated transformations and steady directions.
5. Recognize that a rotation may have no real eigenvector, motivating complex numbers.

Wed Nov 18 — Characteristic equation, 2×2 eigenproblems, and complex numbers

Lecture 36 page

Reading: Strang §6.1, characteristic equation and the complex-eigenvalue (90° rotation) example.

Note: Meets the explicit hand-calculation and complex-number CLOs. The same free §6.1 file is used Nov 16–20.

By the end of class, students should be able to:

1. Interpret $a + bi$ algebraically and as a point/vector in the complex plane.
2. Perform the basic complex arithmetic needed for eigenvalue calculations.
3. Form and solve $\det(A - \lambda I) = 0$ for a 2×2 matrix.
4. Find corresponding eigenvectors.
5. Check real or complex eigenpairs by substitution.

Fri Nov 20 — Why did the system behave that way?

Lecture 37 page

Reading: Strang §6.1, Lab 4, and Lachniet Ch. 5 (eigenvalues/eigenvectors).

Note: Normal modes are an application/demo, not a differential-equations derivation. Students then use software in Lab 4.

By the end of class, students should be able to:

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify $Av \approx \lambda v$ numerically.
3. Use dominant eigenbehavior to explain a simulated system.
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.

Week 15

Mon Nov 30 — When $Ax = b$ has no exact solution

Lecture 38 page

Reading: Boyd 12.1

By the end of class, students should be able to:

1. Recognize an inconsistent overdetermined system.
2. Formulate the least-squares objective $\min_x \|Ax - b\|$.
3. Interpret the fitted vector as the closest attainable vector.
4. Interpret the residual geometrically.
5. Distinguish an exact solution from a best approximate solution.

Wed Dec 2 — Least squares, data fitting, and positive definiteness

Lecture 39 page

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, Lab 5, and Lachniet Ch. 2 (least-squares / data-fitting exercises).

Note: Use a small real dataset. Positive definiteness is from Boyd and class notes, not later Strang sections.

By the end of class, students should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of A.
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. Recognize that $A^\mathsf{T} A$ is symmetric positive semidefinite, explain $x^\mathsf{T} A^\mathsf{T} A x = \|Ax\|^2$, and recognize that $A^\mathsf{T} A$ is positive definite when A has linearly independent columns.

Fri Dec 4 — SVD, low-rank approximation, and capstone launch

Lecture 40 page

Reading: Selected slides from the Strang/MIT SVD slides (not all 17 pages), capstone launch, Lachniet Ch. 5 (SVD) and Ch. 7 (image compression).

By the end of class, students should be able to:

1. Interpret $A = U\Sigma V^\mathsf{T}$ conceptually.
2. Construct a rank-k approximation computationally.
3. Compare singular-value decay for an image/frame and a frame-difference matrix.
4. Encode three frames either independently or as F_1, D_2, D_3.
5. Compare reconstruction error and storage for the two approaches.
6. Formulate a capstone question involving accuracy, storage, runtime, and resource use.

Week 16

Mon Dec 7 — Capstone studio

Lecture 41 page

Reading: Capstone (no new mathematics)

By the end of class, students should be able to:

1. Implement and validate a low-rank computational strategy.
2. Measure approximation error, storage, and runtime.
3. Separate one-time factorization cost from repeated-use cost.
4. Compare theoretical operation/storage counts with empirical measurements.
5. Debug and refine an analysis collaboratively while retaining individual understanding.

Wed Dec 9 — Capstone presentations / gallery walk

Lecture 42 page

Reading: Capstone presentations

Note: Pair lightning talks or gallery walk.

By the end of class, students should be able to:

1. Explain the problem and why SVD or another linear-algebra tool was appropriate.
2. Present quantitative evidence about accuracy and cost.
3. Defend a choice of rank or representation.
4. Evaluate classmates' methods constructively.
5. Communicate the practical value and limitations of the result.

Fri Dec 11 — Cumulative final review

Lecture 43 page

Reading: Cumulative review

Note: No routine quiz or new content.

By the end of class, students should be able to:

1. Connect vectors, systems, transformations, dynamics, eigenvectors, least squares, and SVD.
2. Select an appropriate method for a new situation.
3. Decide what should be done by hand and what should be delegated to software.
4. Interpret mathematical and computational results.
5. Create a targeted final-exam study plan.